GUANGZE DAN

PhD Researcher · New Energy Science & Engineering · Interaction Design & Creative Technology

Wuhan, China · +86 180 8389 3420 · Guangze.Dan@alumni.york.ac.uk · guangze-dan.github.io

RESEARCH PROFILE

I design interactive systems that make invisible infrastructure legible to the people who maintain and live with it. My work moves between engineering data, human-centred AI, and immersive media: from a YOLO-based road inspection system at York, to practice-as-research on invisible labour at UCL, to a WebGL testbed for subsurface energy storage at CUG. I am interested in how interfaces can help non-experts trust, correct, and reason with complex data — and how immersive experiences can make infrastructures that are underground, distributed, or algorithmic feel tangible.

EDUCATION

PhD, New Energy Science and Engineering · China University of Geosciences (Wuhan) · 2026–present

- Research: compressed air energy storage (CAES) underground gas storage; large-scale physical model tests; distributed fibre-optic and acoustic-emission monitoring; 3D data visualisation.

MSc, Digital Media · University College London (UCL) · 2024–2025

- Final project: *Animating the Invisible* — a practice-as-research project on invisible labour in content moderation, comprising two stop-motion films and an accompanying research paper.

BSc (Hons), Interactive Media · University of York · 2021–2024

- Upper second-class honours (2:1). Individual Project supervised by Prof. Florian Block.
- Selected modules: Computational Thinking; Developing Interactive Media (72); Individual Project (71); UX Design (64); Esports (66).

RESEARCH & SELECTED PROJECTS

Road Crack Detection System · York Individual Project · 2024 · Supervisor: Prof. Florian Block

- Built a full-stack inspection system for road maintenance teams, improving YOLO detection accuracy and inference speed.
- Designed the workflow from image upload and real-time annotation to automated defect reports, with a front-end aimed at non-expert users.
- Focus: making model outputs trustworthy and correctable by the people who act on them.

Animating the Invisible · UCL MSc final project · 2025

- Practice-as-research project comprising two stop-motion films, *ECHO* and *Render CHAOs*, and an accompanying research paper.
- Examines invisible labour in content moderation and algorithmic governance through material, frame-by-frame animation.

CAES Underground Gas Storage 3D Testbed · China University of Geosciences (Wuhan) · 2025–present

- Building a WebGL-based 3D testbed that turns large-scale physical model tests of subsurface storage into an explorable scene.
- Integrates cyclic loading parameters, fracture configurations, and monitoring data (distributed fibre optics, acoustic emission) into an interactive visual interface.
- Supports research on stability, sealing performance, and reliability of lined rock caverns and abandoned-mine storage.

Yorkshire BIM Visualisation & Real-Time Animation System · Yunnan Transportation Planning and Design Institute · 2024

- Developed a BIM-to-UE5 visualisation pipeline and a custom real-time animation editor for architectural and engineering scenes.
- Produced virtual cinematography and post-production sequences for client-facing presentations.

TECHNICAL SKILLS

Interaction & product design: Figma, Adobe XD, Axure, Photoshop, Illustrator; user research, information architecture, prototyping, usability testing, accessibility, design systems.

Front-end development: HTML5, CSS3, JavaScript/TypeScript, React, Vue, Tailwind CSS, REST APIs, Git.

3D & media: Unreal Engine 5, Unity, Blender, 3ds Max, Maya, BIM, real-time rendering, virtual production, After Effects, Premiere, DaVinci Resolve.

AI & data: YOLO/computer vision, LangChain, RAG/GraphRAG, knowledge graphs, Python, MATLAB.

Engineering & sensing: COMSOL Multiphysics, FLAC3D/FLAC2D, ABAQUS, ANSYS; distributed fibre-optic sensing, acoustic emission, MQTT sensor networks.

EXPERIENCE

Research Assistant · China University of Geosciences (Wuhan) · 2025–present

- CAES physical model platform; sensor integration; 3D animation and visualisation of experimental systems.

Industrial & Interaction Design Tutor · New Oriental SFK Art & Design Education, Wuhan · 2026–present

- Teaches design thinking, user research, physical computing, and AIGC workflows; mentors students from concept to high-fidelity prototype.

LANGUAGES & REFERENCES

Chinese (native) · English (IELTS 7.0). References available on request: Prof. Florian Block, Prof. Jonathan Hook, Dr Abigail Evans, Alessandro Altavilla, Sanjit Samaddar.